

Protocol

OR/M005

December 2021

Spinal Cord Compression Pathway for patients known to have Malignancy

In-patients with a neurological deficit and suspicion of MSCC (malignant spinal cord compression), start Dexamethasone 8mg BD stat with PPI cover in patients with no signs of sepsis. Get MRI whole spine unless contra-indicated within 24 hours of presentation

MRI is available at the weekend at all DGHs except QEH, Kings Lynn (Sat 9-12 only)

1 MSCC Coordinator (MSCCC)

CUH = Oncology SpR on call

DGH = Local Oncology/AOS team unless weekend; contact CUH Oncol SpR on call

Please ensure all patients have had Oncology input (and prognosis clearly documented) prior to Neurosurgical referral, which should be made by the MSCCC or local clinician to whom the MSCCC has delegated the referral duty.

2 DGH Pathway

WSH/Bedford/Kings Lynn will have already had AOS input and MSCCC will only be contacting neurosurgery for surgical decision; Neurosurgery to take call – patients in whom surgery not considered to remain under appropriate local team

- Lister AOS+RT is at Mt Vernon; They should have discussed with local Oncology team (Mt Vernon) and MSCCC will only be contacting neurosurgery for decision on surgery; Neurosurgery to take Call– see above
- Hinch/Peterborough have AOS and do their RT Mon-Fri in Peterborough; MSCCC only contact neurosurgery for decision on surgery; Neurosurgery to take call– see above
- At weekends we provide RT service; Oncology to take call, discuss with neurosurgery/neurology depending on query

3 Limit contact with neurosurgery to these cases;

- Prognosis >6months (Tokuhashi score)* and
- Unstable spine (SINS score) and/or
- Limited spinal disease and/or
- Previously irradiated area with no scope for further RT

*Note: Tokuhashi score can be overridden by the Oncology team

For patients who do not meet the above criteria and following Oncology review, if still thought appropriate for consideration of surgery then decision making will be made by Consultant to Consultant discussion

If paralysis for >24 hours RT has limited/no benefit but can be used for pain control

RT post-surgery will be considered once wound healed and patient discharged home

Patients whose pain or neurological symptoms get worse when they stand or walk, or with a SINS score > 13, should be on bed rest until their images and clinical presentation have been reviewed by a spinal surgeon

4 Radiotherapy;

Radiotherapy recommended dose is 8Gy single fraction

For new diagnosis and patients with good prognosis, 20Gy in 5 can be given

Retreatment dose is usually 16Gy in 4.

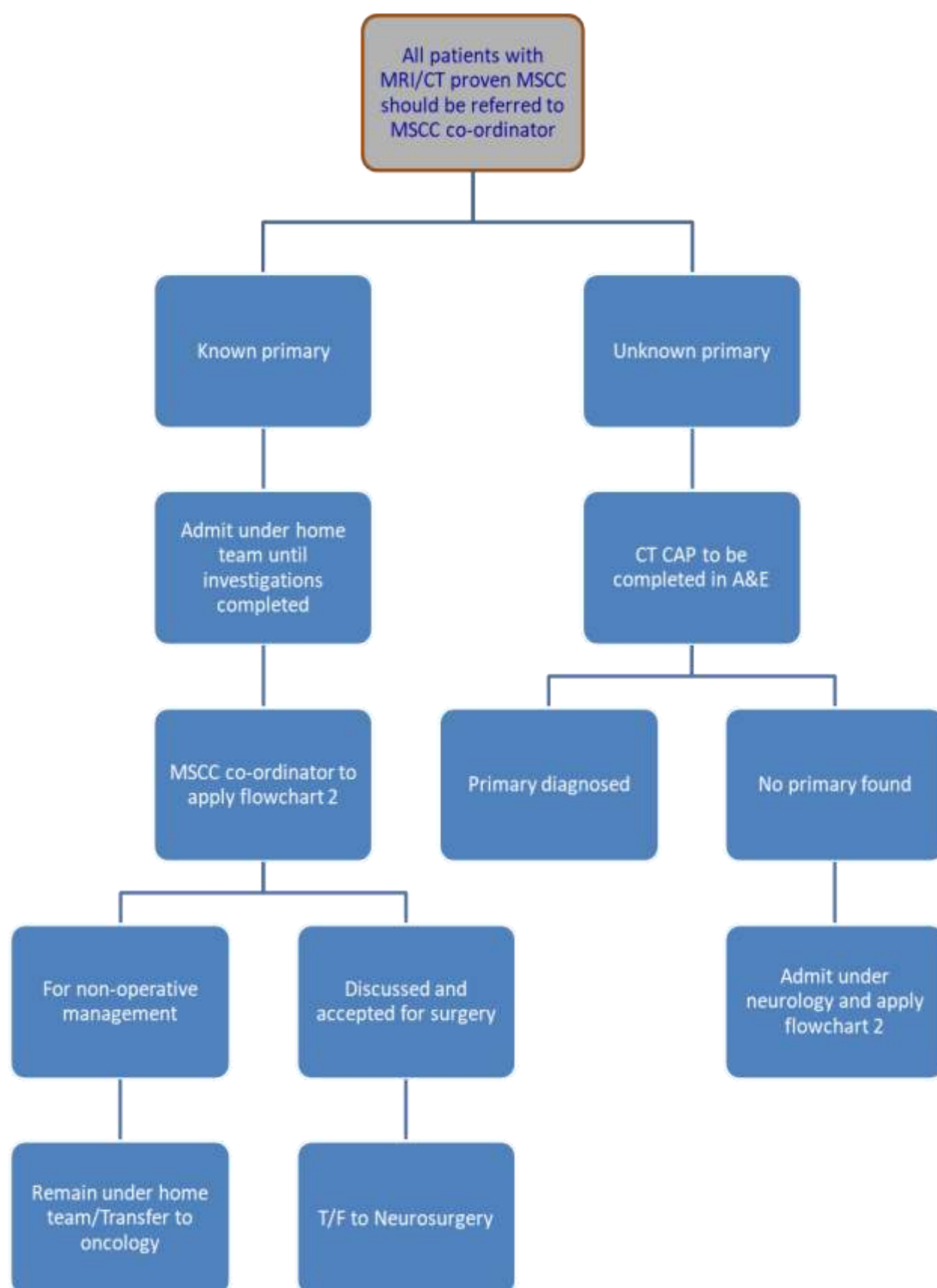
Urgent weekend radiotherapy delivery;

Radiotherapy Dept is open 9-12 with radiotherapy treatment delivered 10-12 sat + sun.

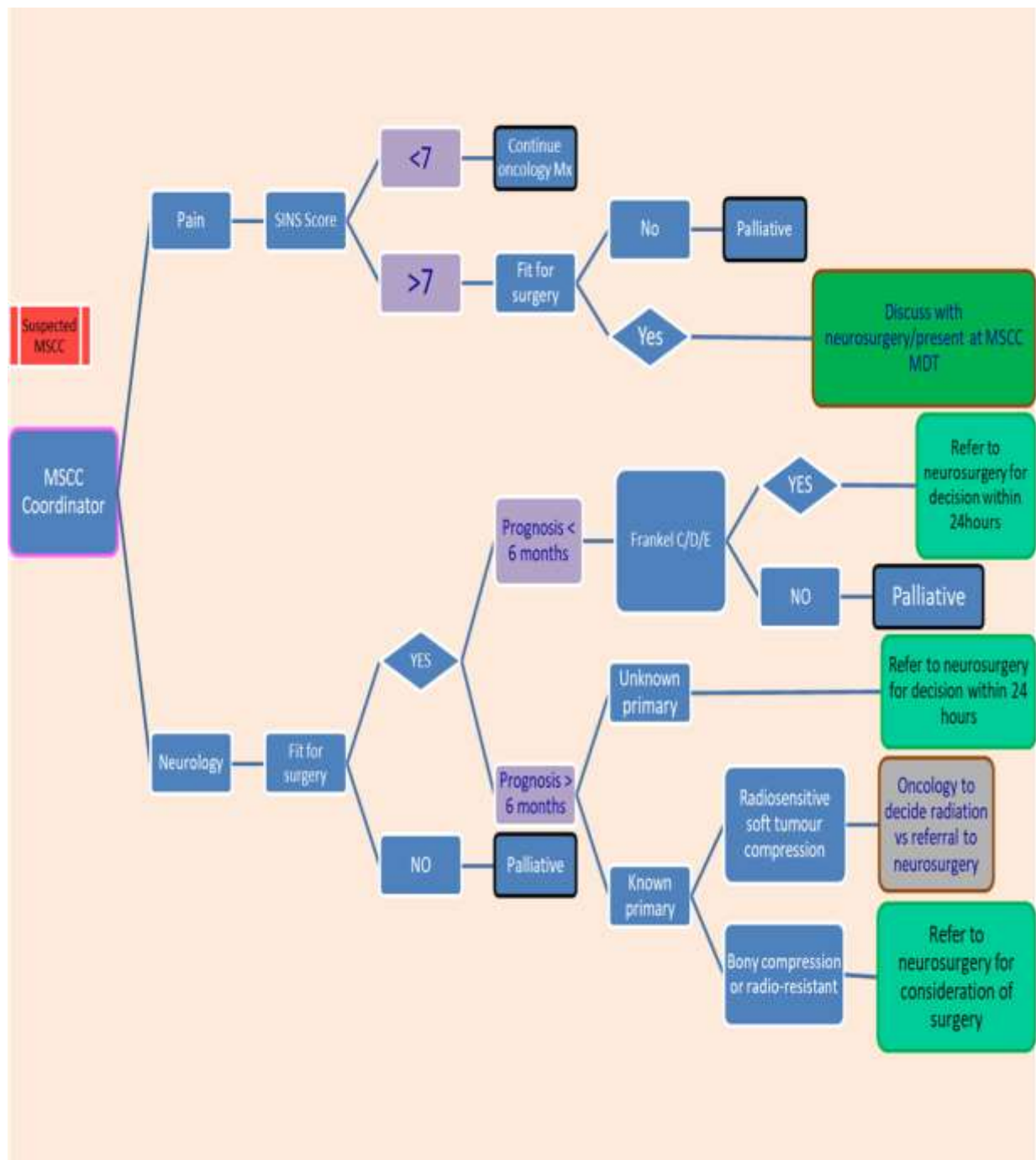
Radiotherapy at the weekend is prescribed as 8Gy to depth of 6cm. 998MU of 10MV x-rays should be given at 100cm FSD. See preset on call RT prescription.

If you wish to deliver 20Gy in 5 fractions, prescribe 4Gy at the weekend, and the plan will be calculated on Monday, to deliver the remainder of the RT. 450MU at 100cm FSD will be delivered. Please state the depth you want it prescribed for on the prescription, for use in the calculation of the subsequent calculations.

Pls state clearly on the radiotherapy prescription that the plan is 20Gy in 5 fractions.



MSCC Flowchart 2



Tokuhashi score; (Prognosis guide)

Characteristic	Score
General condition (performance status [PS])	
Poor (PS 10%-40%)	0
Moderate (PS 50%-70%)	1
Good (PS 80%-100%)	2
No. of extraspinal bone metastases foci	
≥3	0
1-2	1
0	2
No. of metastases in the vertebral body	
≥3	0
1-2	1
0	2
Metastases to the major internal organs	
Unremovable	0
Removable	1
No metastases	2
Primary site of the cancer	
Lung, osteosarcoma, stomach, bladder, esophagus, pancreas	0
Liver, gallbladder, unidentified	1
Others	2
Kidney, uterus	3
Rectum	4
Thyroid, breast, prostate, carcinoid tumor	5
Palsy	
Complete (Frankel A, B)	0
Incomplete (Frankel C, D)	1
None (Frankel E)	2

^aCriteria of predicted prognosis: Total score (TS) 0-8, <6 mo; TS 9-11, 6-12 mo; TS 12-15, ≥1 y.

SINS Score (Spinal Stability score)	
SINS component	Score
Location	
Junctional (occiput–C2, C7–T2, T11–L1, L5–S1)	3
Mobile spine (C3–6, L2–4)	2
Semirigid (T3–10)	1
Rigid (S2–5)	0
Pain*	
Yes	3
Occasional pain but not mechanical	2
Pain-free lesion	0
Bone lesion	
Lytic	2
Mixed (lytic/blastic)	1
Blastic	0
Spinal alignment	
Subluxation/translation present	4
<i>De novo</i> deformity (kyphosis/scoliosis)	2
Normal alignment	0
Vertebral body collapse	
>50% collapse	3
<50% collapse	2
No collapse with >50% body involved	1
None of the above	0
Posterolateral involvement of the spinal elements[†]	
Bilateral	3
Unilateral	1
None of the above	0
Criteria of instability. Total score (TS) 0–6 : stable spine, TS 7–12 : potential unstable spine, TS 13–18 : unstable spine. Recommendation : TS ≥7, consider surgical intervention. *Pain improvement with recumbency and/or pain with movement/loading of the spine, [†] Facet, pedicle, or costovertebral joint fracture or replacement with tumor	

5 Monitoring compliance with the effectiveness of this document

- a) Process for monitoring compliance and effectiveness: Review of Safety Learning Events, as recorded on the Trust QGIS reporting system, for non-compliance and the results presented to the Patient Safety & Governance group - the minutes of these meeting are retained for a minimum of 5 years.
- b) Standards/Key Performance Indicators: This process forms part of a quality system accredited to International Standard BS EN ISO 9001:2015. The effectiveness of the process will be monitored in accordance with the methods given in the quality manual ([QM1.00](#)).

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